

2. Handle more complicated errors;
3. Keep ASR transcripts unmodified as much as possible from the LLM outputs;
4. Avoid undesired leading or tailing text from the generated completions, such as “Here is the corrected transcript” or “We corrected the speakers for these words”;
5. Use smaller and cheaper LLMs.

To finetune the LLM, we build our training data as a collection of prompt-completion pairs. First, for each utterance, we run the ASR model and the speaker diarization system on it, and apply the orchestration module as shown in Fig. 2. This will produce the hypothesis word sequence $\mathbf{w}^{hyp}$ and hypothesis speaker sequence $\mathbf{s}^{hyp}$. From the ground truth annotations of this utterance, we build the reference word sequence $\mathbf{w}^{ref}$ and the reference speaker sequence $\mathbf{s}^{ref}$. Given these four sequences, we can build the prompts and completions in our training data with three different flavors, as introduced below.

3.5.1 FLAVOR 1: HYP2ORA

The first flavor is named *hypothesis-to-oracle*, or simply *hyp2ora*. In this flavor, we apply the Transcript-Preserving Speaker Transfer algorithm from Section 3.4 by treating reference sequences as source and hypothesis sequences as target:

$$\mathbf{s}^{ora} = \text{TPST}(\mathbf{w}^{ref}, \mathbf{s}^{ref}, \mathbf{w}^{hyp}, \mathbf{s}^{hyp}), \quad (1)$$

where the output $\mathbf{s}^{ora}$ is the **oracle hypothesis speakers** transferred from the reference sequences. With $\mathbf{s}^{ora}$, the prompts and completions in our training data are created as below:

- *Prompts*: The text representation of $\mathbf{w}^{hyp}$ and $\mathbf{s}^{hyp}$, with segmentation, and optionally prefix and suffix.
- *Completions*: The text representation of $\mathbf{w}^{hyp}$ and $\mathbf{s}^{ora}$, with segmentation, and optionally suffix.

3.5.2 FLAVOR 2: DEG2REF

The second flavor is named *degraded-to-reference*, or simply *deg2ref*. In this flavor, we apply the Transcript-Preserving Speaker Transfer algorithm from Section 3.4 by treating hypothesis sequences as source and reference sequences as target:

$$\mathbf{s}^{deg} = \text{TPST}(\mathbf{w}^{hyp}, \mathbf{s}^{hyp}, \mathbf{w}^{ref}, \mathbf{s}^{ref}), \quad (2)$$

where the output $\mathbf{s}^{deg}$ is the **degraded reference speakers** transferred from the hypothesis sequences. With $\mathbf{s}^{deg}$, the prompts and completions in our training data are created as below:

- *Prompts*: The text representation of $\mathbf{w}^{ref}$ and $\mathbf{s}^{deg}$, with segmentation, and optionally prefix and suffix.
- *Completions*: The text representation of $\mathbf{w}^{ref}$ and $\mathbf{s}^{ref}$, with segmentation, and optionally suffix.

3.5.3 FLAVOR 3: MIXED

The third flavor named *mixed* is simply the union of the prompts and completions from the previous two flavors. When building training batches, prompt-completion pairs from the two flavors are interleaved.

Note that for all three flavors, it is critical for the prompt and completion to use the same word sequence with different speaker sequences. This helps the LLM to focus on correcting the speaker labels without modifying the ASR transcripts.

4 EXPERIMENTS

4.1 DATASETS

To finetune the LLM, we use the training subset of the Fisher corpus [40], which consists of 1,920 hours of 11,527 conversations. The same train-test split of the Fisher dataset has been used in many previous works [8, 17, 35, 41]

For evaluation, we use the testing subset of the Fisher corpus [40], as well as the testing subset of Callhome American English data [42]. The Fisher testing subset consists of 28.7 hours of 172 conversations³. The Callhome American English testing subset consists of 1.7 hours of 20 conversations. Both datasets are in the telephone speech domain, and all conversations have 2 speakers.

4.2 METRICS

To evaluate the diarization performance, we use two metrics: the Word Diarization Error Rate (WDER) [25] and the concatenated minimum-permutation word error rate (cpWER) [43]. To briefly recap, WDER is defined as:

$$\text{WDER} = \frac{S_{\text{IS}} + C_{\text{IS}}}{S + C}, \quad (3)$$

where,

1. S_{IS} is the number of ASR Substitutions with Incorrect Speaker tokens.
2. C_{IS} is the number of Correct ASR words with Incorrect Speaker tokens.
3. S is the number of ASR substitutions.
4. C is the number of Correct ASR words.

And cpWER is computed as follows:

1. Concatenate all transcripts of each speaker for both reference and hypothesis.
2. Compute the WER between the reference and all possible speaker permutations of the hypothesis.
3. Pick the lowest WER among all these permutations, which is assumed to be the best permutation.

All three metrics reported in this paper (WER, WDER, and cpWER) are micro metrics, i.e. both numerators and denominators are aggregated on the entire dataset.

4.3 MODELS

For the ASR model in Fig. 2, we use a universal speech model (USM) [44] with 600 million parameters trained with the RNN-T loss [45]. For the speaker diarization model in Fig. 2, we use the turn-to-diarize system [7] with a multi-stage clustering setup [17] in our experiments, which is capable of diarizing hours of audio recordings in real time on a mobile device [24]. The number of speakers is unknown (from 1 to ∞) to the speaker diarization system in all of our experiments. Specifically, for the Fisher training set, the Fisher testing set, and the Callhome testing set, all utterances have a ground truth number of speakers equal to two, but our turn-to-diarize system may predict any number of hypothesis speakers. The histogram of number of hypothesis speakers predicted by turn-to-diarize on the Fisher training set is shown in Figure 3. Although our experiments are based on the USM + turn-to-diarize setup, we would like to point out that the proposed framework is very generic and should work with other ASR or speaker diarization systems as well, such as variants of end-to-end speaker diarization models [20, 21, 22, 23].

For the LLM in Fig. 2, we experiment with the PaLM 2-S model (“text-bison” model in Google Cloud API) and the PaLM 2-L model (“text-unicorn” model in Google Cloud API) [46]. We use

³<https://github.com/google/speaker-id/blob/master/publications/ScdLoss/eval/fisher.txt>

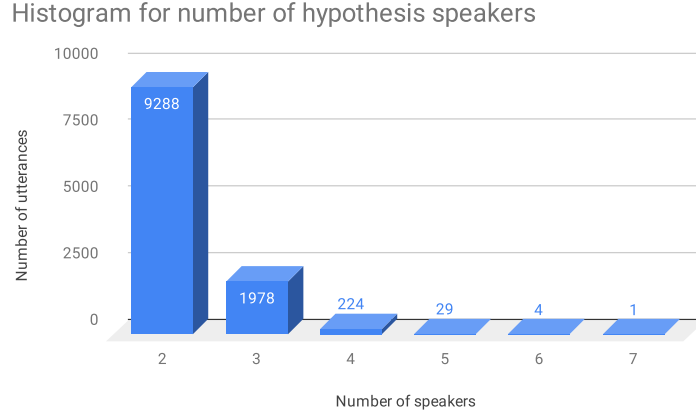


Figure 3: The histogram for the number of hypothesis speakers predicted by the turn-to-diarize system on the Fisher training set. Note that the ground truth number of speakers is always two on the Fisher dataset, but we do not constrain the number of speakers for the turn-to-diarize system.

the PaLM 2-S model as our foundation model, and finetune it on the dataset described in Section 4.1 with data processing steps described in Section 3.5. This model uses a sentence piece model (SPM) of 256k tokens as its tokenizer [47]. During finetuning, we limit the LLM input size by 4,096 tokens, and segment our training and testing data accordingly. The PaLM 2-L model will only be used for zero-shot and one-shot experiments, as described in Section 4.4. We also experimented with open source models such as Llama 2 [48] and Llama 3. Model details on how we finetune these models are provided in Appendix A.

In our prompt builder module, we use an empty prompt prefix, and a 5-character prompt suffix “`_-->_`” (note the two spaces around the arrow). For the completions in our training data, we use a 6-character completion suffix “`_ [eod]`” (short for “end of document”; note the leading space). After processing the training data with the prompt builder module, we result in 13,430 prompt-completion pairs for training in total. The average length of a prompt is 2,371 SPM tokens, and the average length of a completion is 2,329 tokens. The LLM is trained for 1,200 steps with a batch size of 16.

4.4 ZERO-SHOT AND ONE-SHOT BASELINES

Apart from finetuning the PaLM 2-S model on the speaker diarization task, we also experiment with directly using the PaLM 2-S and PaLM 2-L models on the speaker diarization task without finetuning. This is more similar to the example we demonstrated in Section 2.

For the zero-shot setup, we use a prompt prefix that contains an instruction describing the task, as shown below.

Prompt prefix for zero-shot

In the speaker diarization transcript below, some words are potentially misplaced.
Please correct those words and move them to the right speaker.
Directly show the corrected transcript without explaining what changes were made or why you made those changes.\n

For the one-shot setup, the prompt prefix contains both the instruction describing the task, and also a small example, as shown below.